Class 12 Lab Session

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Section 1. Proportion of G/G in a population

Downloaded a CSV file from Ensemble < https://www.ensembl.org/Homo_sapiens/Variation/Sample?db=core;r=17:39894595-39895595;v=rs8067378;vdb=variation;vf=959672880#373531_tablePanel >

Here we read this CSV file

```
mxl <- read.csv("373531-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
head(mxl)
```

```
  Sample..Male.Female.Unknown. Genotype..forward.strand. Population.s. Father
1                  NA19648 (F)                       A|A ALL, AMR, MXL      -
2                  NA19649 (M)                       G|G ALL, AMR, MXL      -
3                  NA19651 (F)                       A|A ALL, AMR, MXL      -
4                  NA19652 (M)                       G|G ALL, AMR, MXL      -
5                  NA19654 (F)                       G|G ALL, AMR, MXL      -
6                  NA19655 (M)                       A|G ALL, AMR, MXL      -
  Mother
1      -
2      -
3      -
4      -
5      -
6      -
```

```r
table(mxl$Genotype..forward.strand.)
```

```
A|A A|G G|A G|G 
 22  21  12   9 
```

```r
table(mxl$Genotype..forward.strand.) / nrow(mxl) * 100
```

```
    A|A     A|G     G|A     G|G 
34.3750 32.8125 18.7500 14.0625 
```

Now let's look at a different population. I picked the GBR.

```r
gbr <- read.csv("373522-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
head(gbr)
```

```
  Sample..Male.Female.Unknown. Genotype..forward.strand. Population.s. Father
1                  HG00096 (M)                       A|A ALL, EUR, GBR      -
2                  HG00097 (F)                       G|A ALL, EUR, GBR      -
3                  HG00099 (F)                       G|G ALL, EUR, GBR      -
4                  HG00100 (F)                       A|A ALL, EUR, GBR      -
5                  HG00101 (M)                       A|A ALL, EUR, GBR      -
6                  HG00102 (F)                       A|A ALL, EUR, GBR      -
  Mother
1      -
2      -
3      -
4      -
5      -
6      -
```

Find proportion of G|G

```r
round(table(gbr$Genotype..forward.strand.) /nrow(gbr) * 100, 2 )
```

```
  A|A   A|G   G|A   G|G 
25.27 18.68 26.37 29.67 
```

This variant that is associated with childhood asthma is more frequent in the GBR population than the MKL population.

Let's now dig into this further.

Section 4: Population Scale Analysis

> **Q13.** Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

```r
file <- "rs8067378_ENSG00000172057.6.txt"
data <- read.table(file)

colnames(data) <- c("sample", "genotype", "expression")

head(data)
```

```
   sample genotype expression
1 HG00367      A/G   28.96038
2 NA20768      A/G   20.24449
3 HG00361      A/A   31.32628
4 HG00135      A/A   34.11169
5 NA18870      G/G   18.25141
6 NA11993      A/A   32.89721
```

```r
table(data$genotype)
```

```

A/A A/G G/G
108 233 121
```

The sample size for the A/A genotype is 108, the A/G genotype is 233, and G/G genotype is 121.

```r
tapply(data$expression, data$genotype, median)
```

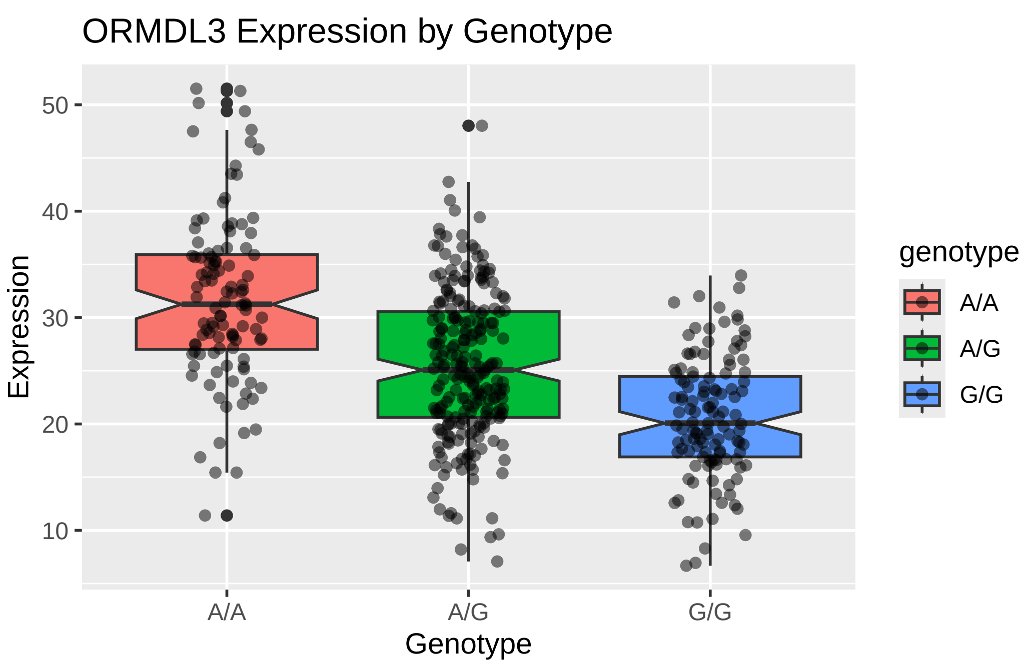
```
     A/A      A/G      G/G
31.24847 25.06486 20.07363
```

The corresponding median expressions are 31.24847 (A/A), 25.06486 (A/G), and 20.07363 (G/G).

> **Q14.** Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

```
library(ggplot2)

ggplot(data,
       aes(x = genotype, y = expression, fill = genotype)) +
       geom_boxplot(notch = TRUE) +
       geom_jitter(width = 0.15, alpha = 0.5) +
  labs(
    title = "ORMDL3 Expression by Genotype",
    x = "Genotype",
    y = "Expression") +
  theme_gray()
```



This boxplot shows that the A/A genotype individuals have higher expression levels of ORMDL3 compared to those with G/G. This means that the rs8067378 SNP affects the expression of the ORMDL3 gene because the G allele is associated with a decrease in gene expression.